FORKAUDIT: HYBRID-LLM OWNERSHIP-TRACE VALIDATION BEYOND OUTPUT EQUALITY

Anonymous authors
Paper under double-blind review

ABSTRACT

Forking many continuations from one prefix creates an ownership question that output equality cannot resolve: agreeing branches may still alias mutable recurrent state or share a defect. We present FORKAUDIT, a phase-aware ownership-trace validator for hybrid KV and recurrent state whose byte-bound lifecycle, storage, tail-COW, transition, call, and semantic receipts fail closed: missing mandatory evidence prevents a pass and leaves the target open.

On one fixed Qwen3.5/H20 stack, six of seven contract targets have complete trace coverage and a passing replay verdict; dispatch passes at Python-call scope with explicitly partial compiled-dispatch coverage. Across 96 KV-by-GDN ownership configurations, every registered semantic observable matches, while transition receipts bind each completed request's 60 mutable GDN tensors as disjoint from bases and peers. All nine preregistered primary faults reach their frozen first gates; when each named gate is separately suppressed, five reach semantics, where tokens catch $0/5$ and exact logits catch $1/5$. Separately, 44 captured-boundary numerical rows pass and 44 wrong-operator controls fail. Process-separated GDN relation reconstruction passes all six phase and both lifecycle checks. A designer–executor-separated campaign freezes five new matched fault pairs before executor preparation: all clean gates pass, every mutant is rejected first by its frozen predicate, four preserve tokens, and three also preserve full logits plus terminal logical KV/GDN. These results establish fixed-stack validation for non-adversarial offline CI at declared observation points, not runtime attestation; faithful producer enumeration remains in the trusted computing base (TCB).

1 INTRODUCTION

Best-of-N decoding, agents, tree search, and document QA fork many continuations from one prefix. Paged KV caches, prefix trees, and scheduling exploit that structure (Kwon et al., 2023; Zheng et al., 2024; Gim et al., 2024; Srivatsa et al., 2025). Hybrid models add mutable recurrent or linear-attention state to read-only KV, so sharing also requires an explicit request-local recurrent view (Yang et al., 2025; Lenz et al., 2025; Pan et al., 2025).

Output equality is only relational: branches can agree while aliasing state, and compared implementations can share a defect. An ownership audit must also witness prefix immutability, phase-conditioned request/base and request/peer relations, tail COW, transition-time rebinding, and call provenance. Likewise, logical payload, cache pages, allocator peaks, process memory, and admitted capacity are distinct memory denominators.

FORKAUDIT partitions document and request state, records page reservations and recurrent rebinding, fingerprints the Python attention callable, and combines a 2×2 ownership factorial with relational, physical-storage, and selected numerical checks. Its executable contract joins a typed phase-indexed schema, fail-closed coverage, and ownership predicates spanning KV and recurrent state. It targets accidental state-management regressions in instrumented offline CI; the capture/replay path is an explicit TCB (Section 3).

The methodological contribution is the executable ownership contract: a target can pass only when phase-complete evidence, pointer-free ownership predicates, and semantic relations all hold for both KV and recurrent state. Output relations alone can miss aliasing, while storage checks without

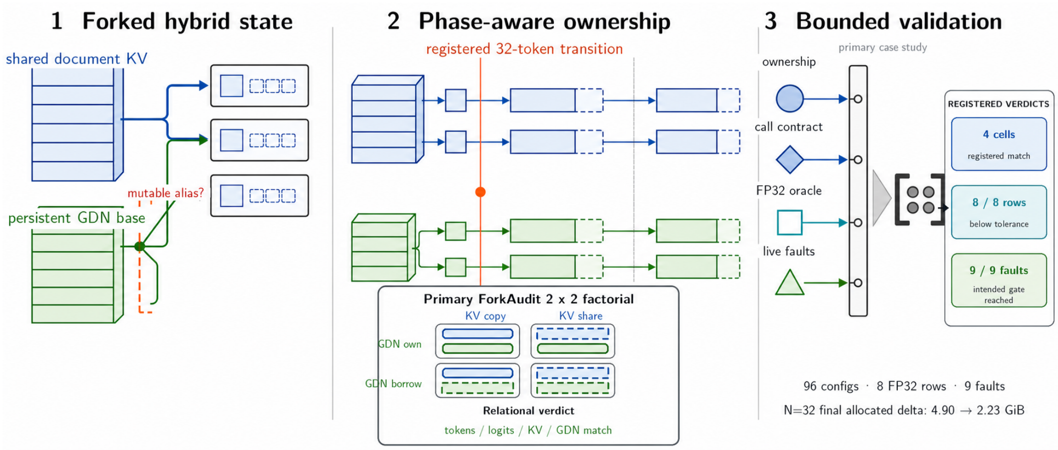

Figure 1: FORKAUDIT in one view. A common prefix forks into requests with distinct KV and recurrent-state lifetimes. Four ownership cells feed private append and recurrent-rebinding checks. The right-hand audit rail maps ownership/call receipts, selected FP32 attention checks, and positive-control faults into registered verdicts. The callouts summarize 96 configurations, selected numerical checks, nine primary live faults, and the $N = 32$ endpoint. This is a protocol schematic; quantitative authority comes from the registered results in Section 5.

lifecycle evidence can miss transition-time rebinding. We instantiate this contract—not a new cache policy or production runtime—on one fixed stack:

- **Fail-closed hybrid-state contract.** Mandatory records cover document KV, private append, GDN bases and rebinding, tail COW, and call contracts; missing records prevent a pass, and coverage remains distinct from the replay verdict.
- **Replayable ownership evidence.** Across 96 configurations, semantic invariance is coupled to registered-point physical isolation. Process-separated receivers derive 1,080 GDN descriptors and 96,660 relations from opaque capture IDs and CUDA-IPC handles while slot enumeration stays in the TCB.
- **Falsification-oriented evaluation.** Captured-boundary numerical controls, nine primary faults, and five designer–executor-separated faults show violations that tokens and even exact logits can miss. Outcomes are reported per frozen fault, not as a population rate.

We instantiate the contract on one fixed Qwen3.5/H20 stack with batch-one, sequential round-major calls; Section 6 states its boundary.

2 BACKGROUND AND RELATED WORK

Paged and prefix-aware inference. PagedAttention, Prompt Cache, RadixAttention, ChunkAttention, Hydragen, and Preble establish block sharing, prefix reuse, shared-prefix computation, and scheduling (Kwon et al., 2023; Gim et al., 2024; Zheng et al., 2024; Ye et al., 2024; Juravsky et al., 2024; Srivatsa et al., 2025). We instead ask what execution evidence demonstrates isolation across coexisting state types.

Hybrid state and cache management. Gated Delta Networks maintain mutable compact state (Yang et al., 2025); Jamba, Marconi, and HYPIC address hybrid architectures or reusable hybrid prefixes (Lenz et al., 2025; Pan et al., 2025; Liu et al., 2026c), while Palu, SqueezeAttention, and HeadKV compress or budget KV (Chang et al., 2025; Wang et al., 2025; Fu et al., 2025). FORKAUDIT validates an ownership boundary rather than replacing those policies; Appendix Table 12 separates published numbers from our controls.

Persistent intermediate residual reuse. CoMem persists a depth-j residual per document token (Liu et al., 2026a;b), motivating our persistent document object; its bounded deployment slice is context, not a new selector or quality–latency claim.

Evaluation and metamorphic testing. SCBench emphasizes cache lifecycles (Li et al., 2025); metamorphic testing, MT-DLComp, and CRADLE motivate preserved relations and independent checks (Chen et al., 2018; Xiao et al., 2022; Pham et al., 2019); DNNMem separates memory denominators (Gao et al., 2020). Appendix Table 9 gives our component-wise boundary.

3 The ForkAudit Contract and Conditional Assurance Boundary

Consider a document prefix D and resident requests $r \in \{1, \dots, N\}$. Let $\mathcal{A}$ be the full-attention layers and $\mathcal{G}$ the recurrent linear-attention layers. For $\ell \in \mathcal{A}$, the state is a read-only set of document pages K_ℓ^D plus a request-private reservation K_ℓ^r. For $\ell \in \mathcal{G}$, it is a persistent prefix base G_ℓ^D plus the request-local recurrent state G_ℓ^r. A logical fork is

$$\mathcal{F}_r(D) = \big(\{K_\ell^D, K_\ell^r\}_{\ell \in \mathcal{A}}, \{G_\ell^D, G_\ell^r\}_{\ell \in \mathcal{G}}\big). \tag{1}$$

Threat model and assurance boundary. Here "audit" means an offline, non-adversarial contract check for debugging and regression testing, not security attestation. The instrumented state-management path may allocate, alias, append, rebind, or dispatch incorrectly, but is not assumed to coherently forge, suppress, or relabel observations. The capture/replay path, mandatory-slot schema, and framework tensor/storage semantics form an explicit TCB, including faithful producer enumeration at every registered phase. Under this TCB, a pass establishes that the registered predicates held at the captured slots and times; it does not validate the TCB or protect against coherent omission/fabrication, malicious-runtime behavior, OS/driver faults, or unobserved transient writes.

A bounded GDN cohort moves descriptor/relation derivation to `spawn`-created processes. The producer supplies opaque capture IDs and CUDA-IPC tensor handles for the frozen slot set; receivers derive descriptors and all-pairs relations and emit no verdict. They receive neither candidate rows/verdicts nor live phase/policy/completion labels, while producer slot enumeration, PyTorch/CUDA-IPC semantics, and paused capture remain in the TCB. We therefore separate mandatory-trace *coverage* from replay *verdicts*.

Seven audit targets may be fully witnessed, partially witnessed, or open:

1. **Frozen identity.** Model, data, query bank, layer geometry, kernel descriptor, and protocol hashes are fixed before execution.
2. **Prefix immutability.** K_ℓ^D and G_ℓ^D have identical digests before and after all continuations.
3. **Private ownership.** KV reservations of distinct shared-document requests are disjoint in the common arena; independent materialized arenas have disjoint storage. At setup, a GDN request either owns a materialized base or holds an explicitly read-only persistent-base alias. At the primary factorial's registered transition, all 60 mutable tensors must be disjoint from the base and every peer.
4. **Tail-safe append.** If the last document page is partial, its valid tokens are copied once to a private page before a request appends.
5. **Dispatch provenance.** Both controls invoke the same recorded Python callable and query/mask/position/append contracts; fallback and full-KV concatenation are forbidden. Compiled binaries and autotuning are not fingerprinted.
6. **Cross-arm equivalence.** For the same query and schedule, the materialized and shared ownership arms must have exactly equal generated tokens, full-vocabulary logits, final logical KV, and request recurrent state.
7. **Cross-N prefix consistency.** For each query observed at two or more fan-outs, its result at every larger fan-out must match its smallest-N result.

Targets 2–5 are pre-output trace obligations; targets 6–7 are relational semantic checks, so matching outputs is insufficient. The nine controls stop at frozen first gates before post-injection semantics: bounded localization, not unseen-fault coverage. Separate executions suppress only the named gate and record the next outcome under frozen precedence; executions are not pooled into a rate.

Conditional trace-validity statement. Let Σ be the frozen case specification, E an execution, and $\tau = C_\Sigma(E)$ the producer trace. For target i, let $M_i(\Sigma)$ be its mandatory record slots and Φ_i its registered replay predicates. Coverage is *complete* only when every slot in $M_i(\Sigma)$ has exactly one schema-valid, byte-bound record. We define

$$\mathrm{Pass}_i(\tau) \iff \left[\mathrm{Coverage}_i(\tau) = \mathrm{complete}\right] \wedge \mathrm{Bind}_\Sigma(\tau) \wedge \bigwedge_{\phi \in \Phi_i} \phi(\tau). \tag{2}$$

If capture is honest and mandatory-event-complete, and manifest binding, digests, and replay are correct, $\mathrm{Pass}_i(\tau)$ implies that every registered predicate held at its captured observation point. A missing or modified mandatory record, or a represented predicate violation, cannot produce a pass. This statement is trace-relative: it provides no conclusion about coherent omission or fabrication, transient writes restored between observations, unrecorded compiled dispatch, common-mode semantics, or unseen-fault error rates.

The out-of-process cohort is a separate supporting check on a bounded subset of GDN storage predicates; it does not change this conditional statement for the complete primary-factorial producer trace.

Minimal adoption workflow and decision rule (normative). ForkAudit (1) freezes code/model/data/protocol/runtime roles before outputs; (2) captures setup, transition, and final events with canonical digests, ranges, ordered receipts, outputs, and allocator snapshots; (3) replays every applicable predicate, labeling coverage complete, partial, or open separately from pass/fail/not-evaluated; and (4), when sensitivity is claimed, freezes the target-to-gate map and retains all outcomes. Here terminal identity is complete and passing, while compiled dispatch/autotuning coverage is partial. Appendix C and Table 16 give the dependency matrix, pointer-free schema, and record obligations.

"Exact" means equality of a canonical serialization: contiguous CPU-FP32 bytes with bound shape/dtype, or the specified logical-KV/GDN digest; it does not claim bitwise comparison of an unarchived device tensor.

Targets 6–7 expose layout- or N-dependent changes in recorded observables. The factorial also captures one statically selected rank-specific attention row's post-RoPE query, K/V, positions, mask, and scale for dense IEEE-FP32 comparison. A candidate-import-free NumPy FP32 oracle evaluates four frozen GDN transitions after native BF16 q/k normalization. Preregistration binds that boundary, $1/\sqrt{128}$ query scale, rows, tolerances, and seeded faults. These checks cover selected operator boundaries, not upstream construction, normalization, every GDN layer, or the end-to-end model.

Figure 2 maps the four primary-factorial cells and their lifecycle; allocator and witness cells are rebuilt separately so guards do not retain tensors during measurement.

4 Trace-Validation Protocol

4.1 Frozen case-study geometry

Primary feasibility uses one fixed Qwen3.5-35B-A3B configuration (Qwen Team, 2026a;b): ten vLLM-0.26 Triton full-attention layers with BF16 ("Q16") KV and 128-token pages, plus thirty Transformers-torch GDN layers. Eight H20-3e ranks run one book each with batch-one sequential calls on one CUDA stream/rank. The ledger binds the Python callable, not the compiled binary; each GDN digest covers 60 tensors.

Each distinct PG-19 book (Rae et al., 2020) supplies a 4,095-token document and 32-token registered transition. Four cells evaluate $N \in \{1, 8, 32\}$ and eight argmax outputs; the query plus seven fed-back outputs give 39 state-appended tokens and exercise COW on the 127-token final page. Appendix Table 5 separates all supporting cohorts.

4.2 Audit sequence

Each primary-factorial cell binds identity, rebuilds allocator and witness executions, records setup–transition–final ownership, and compares tokens, logits, logical KV, and GDN state. Replay adds fixed-axis projections, one dense-FP32 sidecar/rank, and matched live faults.

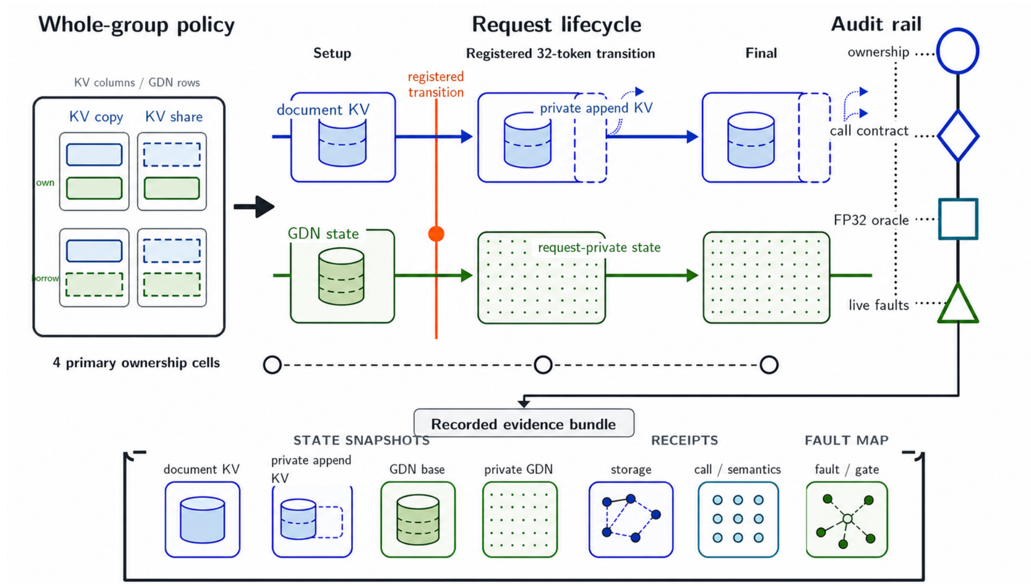

Figure 2: ForkAudit's ownership factorial. The two left axes choose one KV and one GDN setup policy for the request group: append tails are always private, and a borrowed GDN base becomes private at the registered 32-token transition. The dashed lifecycle spine marks setup, that transition, and final capture. Left-to-right, the bracketed bottom row records document KV, private append, persistent GDN, request-private GDN, storage binding, call/semantic, and fault-to-gate evidence. Offline replay validates that bundle and reports coverage separately from each verdict. This mapped schematic is not experimental evidence.

A separate PDF-only designer freeze supplies five executor-separated pairs. Aggregation requires every byte binding; Appendix Table 6 states target strength.

4.3 CONTROLS, SCHEDULES, AND ORACLES

The primary factorial uses sequential round-major steps on one CUDA stream/rank. Full logits use shape/dtype-bound CPU-FP32 bytes; KV, GDN, bases, calls, storage, and allocator records use typed digests. Detached replay cannot reconstruct unarchived tensors or recapture the producer. A preregistered NumPy oracle checks four GDN layers after native q/k normalization against four wrong recurrences.

Supporting protocols move GDN reconstruction to a spawn-created CUDA-IPC observer and record CUDA-event-bracketed model-call intervals around cancel, scrub, reclaim, and replacement. Appendix Table 5 fixes their geometry and authorization.

4.4 MEMORY DENOMINATORS

We separate logical state, pages, synchronized allocator bytes, and unmeasured service capacity. Deployment "Store" is instead the union of tensor byte ranges owned by a retained document entry; it excludes metadata, pools, NVML deltas, and admission capacity.

4.5 GOVERNANCE AND LIVE MUTATION CHECKS

Pre-output ledgers bind code, model, data, queries, and protocol. Nine fresh matched controls pair with live mutations spanning allocation/binding, COW, GDN aliasing, positions, masks, callable identity, and paged-view type; each mutant first stops at its predeclared gate, establishing localization, not a rate; the primary all-gates-on campaign records no post-injection token, logit, logical-KV, or GDN digest. In a separate preregistered 18-case matrix, only that gate is suppressed. The precedence then records another FORKAUDIT gate, an allowlisted production assertion, the exact M8 payload sentinel, or completed-horizon token and FP32 logit equality. Unallowlisted exceptions are invalid; the sentinel is not a production detector.

The designer–executor-separated campaign's designer receives only the prior PDF—not source, evidence, earlier faults, reviews, or state—and freezes five new definitions, loci, payloads, clean gates, precedence, and first predicates before executor preparation or candidate output. Every fault runs all gates; selective reruns and post-output payload tuning are forbidden.

5 Results

5.1 Primary fixed-stack trace validation

Targets 1–4 and 6–7 have complete mandatory-trace coverage and a passing replay verdict for the frozen primary case study. Dispatch provenance (target 5) passes at the recorded Python-call scope but has partial coverage because no per-call compiled-binary or autotuning-choice binding exists. Thus the audit reports six complete passing targets and one explicitly partial passing target rather than silently upgrading missing dispatch evidence.

5.2 Relational equality holds for every registered observable

Across $8 \times 3 \times 4 = 96$ configurations (192 rebuilt allocator/witness cells), tokens, full logits, final logical KV, final GDN, both fixed-axis projections, and all 288 adjacent-fan-out comparisons match exactly. These semantic relations are paired with physical ownership evidence: borrowed GDN starts as 60 immutable base aliases/request, materialized setup is disjoint, and both rebind at the registered 32-token transition boundary to 60 request-private tensors disjoint from bases and completed peers. Every phase passes; $N = 1$ directly exercises request–base disjointness.

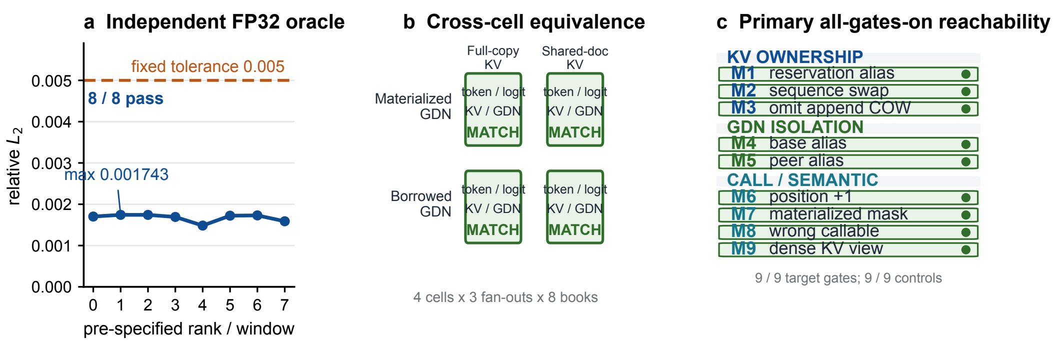

Figure 3: Primary validation. Left: an independently implemented FP32 replay starts from producer-captured post-RoPE inputs, and all selected attention rows beat the pre-fixed tolerance. Center: four ownership cells match on registered observables. Right: a separate primary all-gates-on reference reaches every target gate with passing controls. Together these panels connect selected numerical agreement, cross-cell invariance, and targeted violation localization.

Expanded captured-boundary numerical sweep. The primary dense-FP32 attention oracle obtains relative-L_2 0.001484–0.001743 < 0.005 on eight rows. A preregistered expanded numerical sweep uses two distinct PG-19 inputs and checks 20 attention rows spanning all ten full-attention layers (160 query positions) plus 24 GDN rows spanning 12 of 30 recurrent layers (192 token transitions). All 44 clean rows pass: maximum attention relative-L_2 is 0.001897; maximum GDN output/state relative-L_2 is $0.002073/2.08 \times 10^{-7}$. All 20 attention and 24 GDN seeded wrong-operator controls are rejected. The CPU/NumPy FP32 replay imports no candidate code, but begins from producer-captured post-RoPE attention and post-native-q/k-normalization GDN inputs; Appendix Table 22 fixes this captured-boundary scope.

Table 1: Same-system outcomes for nine designed faults. The primary campaign enables all gates; a separate target-gate-suppression campaign suppresses only the named gate. "Same" is exact and "N/O" an earlier stop. Rows are not a detection rate; M8's injected sentinel is not a production detector.

Fault / target contract	All gates on	With target gate suppressed	Token / FP32 logits
M1 Reservation ownership	target gate	completes	Same / Same
M2 Sequence binding	target gate	completes	Same / Same
M3 Tail COW	target gate	other audit: active-block ownership	N/O / N/O
M4 GDN base isolation	target gate	other audit: peer isolation	N/O / N/O
M5 GDN peer isolation	target gate	completes; logits change ($L_\infty = 0.75$, rel. $L_2 = 0.0314$)	Same / Changed
M6 Positions	target gate	completes	Same / Same
M7 Mask contract	target gate	completes	Same / Same
M8 Callable identity	target gate	injected payload sentinel	N/O / N/O
M9 Paged-KV view	target gate	paired-view production assertion	N/O / N/O

Table 1 adds a prospective same-system comparison. All nine controls pass and no case is invalid. Five suppressed mutants complete: tokens catch none, exact logits catch M5, and M1/M2/M6/M7 retain both. M3/M4 reach redundant audit gates, M9 a production assertion, and M8 its injected sentinel. For the five semantic-complete designed faults, token equality catches $0/5$ and exact logits catch $1/5$, whereas the named ForkAudit gates reject all five in their separate all-gates-on executions. Appendix Table 7 gives the full targeted-control boundary.

Designer–executor-separated fault campaign. Table 2 reports the separately frozen five-pair campaign. Its faults are held out from executor preparation and output inspection, not from FORKAUDIT's predicate vocabulary. Every matched clean case passes complete coverage, byte binding, full-horizon execution, and exact allocator restoration; every mutant also completes and is rejected first by its pre-frozen primary predicate. Four of five preserve the surfaced token sequence. Of the four pairs with comparable call cardinality, three preserve every FP32-logit sidecar byte; the duplicate commit intentionally has one extra recorded call. HF01, HF02, and HF05 also preserve terminal logical KV and GDN state. These are five fixed frozen outcomes, not an estimate over a fault population.

Table 2: Designer–executor-separated PDF-only fault outcomes. A separate designer saw only the prior 24-page PDF and froze all five definitions, loci, payloads, clean requirements, and expected first predicates before executor preparation or candidate output. The faults are held out from executor preparation and outcome inspection, not from the predicate vocabulary. All matched clean cases pass; every mutant completes its registered horizon and is rejected first by the frozen primary predicate. Semantic comparisons are secondary: "Same" means canonical exact equality with the matched clean case, KV/GDN denote terminal logical state, and "N/C" means full-logit call cardinality is not comparable. These five fixed outcomes are not a population detection rate.

Frozen fault	Isolated mutation	First failed predicate	Tokens / logits / KV / GDN
HF01 delayed tail detach	append write precedes the otherwise present tail copy	tail-copy-before-first-write	Same / Same / Same / Same
HF02 inactive-lane scribble	one bit in the unused lane of a shared physical K page	physical-prefix immutability	Same / Same / Same / Same
HF03 duplicate commit	one correct feedback call commits twice; first output discarded	ordered-call cardinality	Same / N/C / Changed / Changed
HF04 effective-scale drift	first registered attention scale is exactly doubled	attention effective scale	Changed / Changed / Changed / Changed
HF05 stale GDN binding token	storage correctly rebinds; lifecycle token remains stale	completed binding-token advance	Same / Same / Same / Same

Same-stack cancellation/reclamation checks match all 16 clean rank–geometry cells and reach the expected gate in all 48 preregistered schedule-fault trials (Tables 18–19), extending the registered lifecycle through cancel, scrub, reclaim, lease change, and replacement under the frozen schedule.

Out-of-process GDN observation. Across two fresh $N = 2$ GDN-policy cells and three phases per cell, one `spawn`-created observer per cell imports live tensor views through PyTorch CUDA IPC. The receiver processes derive 1,080 storage descriptors and all 96,660 pair relations; frozen replay passes 6/6 phase and 2/2 lifecycle verdicts. Both observer processes differ from the producer process, and the worker imports no candidate or model-runtime module and receives no candidate rows, verdicts, or live phase/policy/completion labels. This is process-separated PyTorch/CUDA-IPC GDN observation, not OS/driver ground truth or independent model execution; the producer still enumerates and semantically binds the frozen slots.

Bounded two-stream lifecycle. Two host threads on distinct CUDA streams produce positive full-model-call interval overlap both before cancellation ($260.802\,\mathrm{ms}$) and after quiescent scrub/reclaim/replacement ($95.439\,\mathrm{ms}$). All four tokens and full-logit samples, 20 terminal KV layer pairs, two 60-tensor GDN aggregates, and ownership/lifecycle receipts match the serialized reference. This establishes call-interval overlap and one clean quiescent transition, not simultaneous kernels or arbitrary in-flight cancellation.

Ownership factors move different memory components. At $N = 32$, switching full-copy to shared-document KV reduces the final allocated delta by $2.672\,\mathrm{GiB}$ with materialized GDN setup and $2.661\,\mathrm{GiB}$ with borrowed setup. Borrowing changes allocation timing, not the shared-KV end state: both GDN policies finish at $2.229\,\mathrm{GiB}$. Appendix Table 17 reports the four primary endpoints with their named allocator denominators.

Unpooled Mac motivation context. Across five fresh Apple-M4-Pro processes, CoMem BF16/Q8/Q4 share all selected top-1 decisions yet agree with vanilla dense on only 71.89%. Appendix Table 11 is motivation only, not primary or cross-hardware evidence (Liu et al., 2026a;b).

5.3 Unpooled same-checkpoint deployment and nearest-work context

Table 3 places two unpooled end-to-end blocks beside an independent retained-state cohort. All use the same checkpoint, eight LongBench items, one H20-3e/item, a 4,096-token cap, and greedy decoding; harnesses differ, so timings remain within block. Full-prefix reuse is fastest in our Transformers block; CoMem Q8 records 39.14 F1 at 15.89 MiB and mixed Q4/Q8 records 39.01 F1 at 10.01 MiB. In the separate official SGLang block (Liu et al., 2026c), Prefix Cache is fastest and HYPIC records 39.63 F1, $0.101\,\mathrm{s}$ TTFT, and 17.67 tokens/s; this is a TP=1 adaptation, not its published TP=2 reproduction.

Table 3: Unpooled same-checkpoint H20 context on the eight-item Qasper/2WikiMQA slice. The first block is an in-process Transformers deployment cohort with three timing repeats per item. HYPIC (Liu et al., 2026c) timing/F1 comes from a separate 24-cell TP=1 official-code cohort; its Store values come from an independent receipt-instrumented 8+8-cell cohort. All use Qwen3.5-35B-A3B, one H20-3e per item, a 4,096-token cap, greedy decoding, and at most 32 generated tokens. Timing is interpreted within block; no cross-block speedup, production-capacity, broad-quality, or ForkAudit-ownership claim is made.

Method	Persistent state	Store (MiB)	TTFT (s)	TPOT (ms)	tok/s	F1
Transformers 5.14.1 in-process deployment cohort						
Vanilla dense	–	–	0.649	648.75	1.36	39.42
Full-prefix KV cache	Q16	140.34	0.163	54.11	13.38	39.14
CoMem state	Q8	15.89	0.673	55.44	7.15	39.14
CoMem mixed state	Q4/Q4/Q8	10.01	0.674	54.97	7.19	39.01
CoMem per-layer mixed	mixed	9.74	0.673	55.07	7.19	39.11
CoMem state	Q4	8.41	0.674	55.02	7.19	36.09
HYPIC commit 98147c0 timing/F1; receipt-instrumented Store; SGLang 0.5.14, TP=1						
Official HYPIC code: full recompute	–	–	0.217	49.21	14.77	39.14
Official HYPIC code: prefix cache	Radix prefix	139.53	0.072	50.15	19.07	39.14
HYPIC (transition + seam 8)	segment transition	324.09	0.101	52.13	17.67	39.63

Store is the median retained-document tensor payload: persistent document bytes for CoMem/full-prefix reuse and the union of target-entry-owned physical tensor byte ranges for Prefix Cache/HYPIC. It excludes Python metadata, preallocated pools, NVML/process deltas, and admission capacity. HYPIC timing/F1 uses streaming-client wall-clock from 24 cells; HYPIC Store uses a separate 16-cell externally replayed receipt cohort. Published Qwen3.5 HYPIC panels use TP=2, whereas this fixed same-slice adaptation uses TP=1.

Independent Store cells retain 139.53 MiB/document for Prefix Cache and 324.09 MiB for HYPIC, versus CoMem Q8's 15.89 MiB under the target-owned tensor-range estimand, which excludes metadata, pools, NVML, and capacity. These eight items are context, not a complete LongBench, robustness, or cross-block speedup result.

Further unpooled context. vLLM/SGLang cache controls preserve 8/8 predictions (Table 14); bounded Hydragen/Palu transfers cover one captured layer family (Table 13), not a leaderboard, all-layer result, or combined system.

6 Discussion and Limitations

ForkAudit's verdicts are conditional on the non-adversarial capture TCB in Section 3; the process-separated GDN cohort changes where descriptors and relations are derived, not who enumerates slots. Evidence is limited to one Qwen3.5/H20 stack, bounded schedules, captured-boundary numerical checks, and Python-scope dispatch. The two-stream cohort establishes call-interval rather than kernel overlap and excludes in-flight cancellation. Full synchronous capture and persistence are costly and intended for offline debugging or CI. The designed and designer–executor-separated fault sets establish per-fault sensitivity rather than natural-bug recall, population coverage, or false-positive/false-negative rates. Appendix G gives the exhaustive boundary.

7 Reproducibility Statement

Manifest-first CPU replay covers the primary-factorial package and named replayable supporting cohorts, not every contextual table. The primary package rechecks 536 artifacts, 96 cells, 288 cross-N comparisons, eight attention rows, and nine fault pairs. The target-gate-suppression package rechecks 18 cases and 24 FP32 sidecars; its disclosed wrapper changes only an inherited execution-identifier source and preserves every candidate byte. The out-of-process observer package rechecks six IPC captures, 1,080 receiver-derived rows, 96,660 relations, both observer PIDs, and all six phase/two lifecycle verdicts. The bounded two-stream replay recomputes four full-logit comparisons and both interval overlaps.

The designer–executor-separated package binds the PDF-only author freeze, a lifecycle-amended executor, five clean/mutant pairs, detached clean and pair replays, and 208 terminal files. All five clean gates and expected-first classifications pass; the pre-injection amendment was frozen before all five ranks were rerun in a new output root.

The expanded numerical-sweep package binds 272 sidecars totaling 140,199,936 bytes and rechecks 20 attention and 24 GDN clean rows plus 44 controls. HYPIC binds 4,404 timing artifacts and hash-replays all 16 Store cells. No package independently re-executes the model, enumerates OS/driver allocations, or attests compiled dispatch; Appendix H maps exact artifacts and boundaries.

8 Conclusion

ForkAudit provides fail-closed, replayable hybrid KV/GDN ownership checks beyond output equality. On one fixed Qwen3.5/H20 stack, 96 configurations match every registered semantic observable and bind completed mutable GDN state disjoint from bases and peers; 44 captured-boundary clean rows pass and 44 wrong-operator controls fail. Spawn-created observers derive 1,080 descriptors and 96,660 relations with all six phase and both lifecycle verdicts passing. For five primary designed faults that reach semantics when their named gates are suppressed, tokens and exact logits catch $0/5$ and $1/5$, while separate all-gates-on runs reject all five. Separately frozen PDF-only designer–executor pairs have five passing controls and five mutants rejected first by their frozen predicates; four retain tokens and three retain logits plus terminal logical KV/GDN. These are conditional trace-validity results at declared observation points, not end-to-end or OS/driver ground truth.

References

Chi-Chih Chang, Wei-Cheng Lin, Chien-Yu Lin, Chong-Yan Chen, Yu-Fang Hu, Pei-Shuo Wang, Ning-Chi Huang, Luis Ceze, Mohamed Abdelfattah, and Kai-Chiang

Wu. Palu: KV-cache compression with low-rank projection. In *International Conference on Learning Representations*, volume 2025, pp. 50222–50249, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/7da6e0e00702c60607a6ae05c802ef85-Abstract-Conference.html.

Tsong Yueh Chen, Fei-Ching Kuo, Huai Liu, Pak-Lok Poon, Dave Towey, T. H. Tse, and Zhi Quan Zhou. Metamorphic testing: A review of challenges and opportunities. *ACM Computing Surveys*, 51 (1):4:1–4:27, 2018. doi: 10.1145/3143561. URL https://doi.org/10.1145/3143561.

Yu Fu, Zefan Cai, Abedelkadir Asi, Wayne Xiong, Yue Dong, and Wen Xiao. Not all heads matter: A head-level KV cache compression method with integrated retrieval and reasoning. In *International Conference on Learning Representations*, volume 2025, pp. 99269–99290, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/f649556471416b35e60ae0de7c1e3619-Abstract-Conference.html.

Yanjie Gao, Yu Liu, Hongyu Zhang, Zhengxian Li, Yonghao Zhu, Haoxiang Lin, and Mao Yang. Estimating GPU memory consumption of deep learning models. In *Proceedings of the 28th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, pp. 1342–1352. Association for Computing Machinery, 2020. doi: 10.1145/3368089.3417050. URL https://doi.org/10.1145/3368089.3417050.

In Gim, Guojun Chen, Seung-seob Lee, Nikhil Sarda, Anurag Khandelwal, and Lin Zhong. Prompt cache: Modular attention reuse for low-latency inference. In *Proceedings of Machine Learning and Systems*, volume 6, pp. 325–338, 2024. URL https://proceedings.mlsys.org/paper_files/paper/2024/hash/a66caa1703fe34705a4368c3014c1966-Abstract-Conference.html.

Jordan Juravsky, Bradley Brown, Ryan Ehrlich, Daniel Y. Fu, Christopher Ré, and Azalia Mirhoseini. Hydragen: High-throughput LLM inference with shared prefixes. *arXiv preprint arXiv:2402.05099*, 2024. URL https://arxiv.org/abs/2402.05099.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, pp. 611–626. ACM, 2023. doi: 10.1145/3600006.3613165. URL https://doi.org/10.1145/3600006.3613165.

Barak Lenz, Opher Lieber, Alan Arazi, Amir Bergman, Avshalom Manevich, Barak Peleg, Ben Aviram, Chen Almagor, Clara Fridman, Dan Padnos, Daniel Gissin, Daniel Jannai, Dor Muhlgay, Dor Zimberg, Edden Gerber, Elad Dolev, Eran Krakovsky, Erez Sa, Erez Schwartz, Gal Cohen, Gal Shachaf, Haim Rozenblum, Hofit Bata, Ido Blass, Inbal Magar, Itay Dalmedigos, Jhonathan Osin, Julie Fadlon, Maria Rozman, Matan Danos, Michael Gokhman, Mor Zusman, Naama Gidron, Nir Ratner, Noam Gat, Noam Rozen, Oded Fried, Ohad Leshno, Omer Antverg, Omri Abend, Or Dagan, Orit Cohavi, Raz Alon, Ro'i Belson, Roi Cohen, Rom Gilad, Roman Glozman, Shahar Lev, Shai Shalev-Shwartz, Shaked Meirom, Tal Delbari, Tal Ness, Tomer Asida, Tom Ben Gal, Tom Braude, Uriya Pumerantz, Joshua Cohen, Yonatan Belinkov, Yuval Globerson, Yuval Levy, and Yoav Shoham. Jamba: Hybrid transformer–mamba language models. In *International Conference on Learning Representations*, volume 2025, pp. 67959–67984, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/a9ed43fa31dc8b4a7d7a673d713dcb5f-Abstract-Conference.html.

Yucheng Li, Huiqiang Jiang, Qianhui Wu, Xufang Luo, Surin Ahn, Chengruidong Zhang, Amir Abdi, Dongsheng Li, Jianfeng Gao, Yuqing Yang, and Lili Qiu. SCBench: A KV cache-centric analysis of long-context methods. In *International Conference on Learning Representations*, volume 2025, pp. 66063–66093, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/a540b17fb2295c736d5afd6c507acf66-Abstract-Conference.html.

Hanzuo Liu, Xuan Qi, Chunyu Liu, Haotian Zhong, Yulong Wang, Key, Rayying, Alex Lamb, and Mingyu Gao. CoMem: Reusing transformer depth across queries with persistent intermediate residuals. In *COLM 2026 Workshop on Efficient Reasoning*, 2026a.

Hanzuo Liu, Xuan Qi, Chunyu Liu, Haotian Zhong, Yulong Wang, Rayying, Key, Alex Lamb, and Mingyu Gao. Understanding is done early: A depth division of labor in large language models and its use for unbounded-context memory. *arXiv preprint arXiv:2607.28263*, 2026b. doi: 10.48550/arXiv.2607.28263. URL https://arxiv.org/abs/2607.28263.

Yifei Liu, Juntong Wu, Yang Liu, Junhao Hu, Minghao Li, Xiaoxu Chen, and Weihang Chen. HYPIC: Accelerating hybrid-attention LLM serving with position-independent caching. *arXiv preprint arXiv:2607.01299*, 2026c. doi: 10.48550/arXiv.2607.01299. URL https://arxiv.org/abs/2607.01299.

Rui Pan, Zhuang Wang, Zhen Jia, Can Karakus, Luca Zancato, Tri Dao, Yida Wang, and Ravi Netravali. Marconi: Prefix caching for the era of hybrid LLMs. In *Proceedings of Machine Learning and Systems*, volume 7. MLSys, 2025. URL https://proceedings.mlsys.org/paper_files/paper/2025/hash/7c180af017258d239bac6248d1eb26ac-Abstract-Conference.html.

Hung Viet Pham, Thibaud Lutellier, Weizhen Qi, and Lin Tan. CRADLE: Cross-backend validation to detect and localize bugs in deep learning libraries. In *2019 IEEE/ACM 41st International Conference on Software Engineering (ICSE)*, pp. 1027–1038. IEEE, 2019. doi: 10.1109/ICSE.2019.00107. URL https://doi.org/10.1109/ICSE.2019.00107.

Qwen Team. Qwen3.5: Towards native multimodal agents. Official Qwen technical blog, February 2026a. URL https://qwen.ai/blog?id=qwen3.5.

Qwen Team. Qwen3.5-35B-A3B-Base model card. Hugging Face model repository, 2026b. URL https://huggingface.co/Qwen/Qwen3.5-35B-A3B-Base/blob/main/README.md. Accessed 2026-08-16.

Jack W. Rae, Anna Potapenko, Siddhant M. Jayakumar, Chloe Hillier, and Timothy P. Lillicrap. Compressive transformers for long-range sequence modelling. In *International Conference on Learning Representations*, 2020. URL https://iclr.cc/virtual_2020/poster_SylKikSYDH.html.

Vikranth Srivatsa, Zijian He, Reyna Abhyankar, Dongming Li, and Yiying Zhang. Preble: Efficient distributed prompt scheduling for LLM serving. In *International Conference on Learning Representations*, volume 2025, pp. 37057–37082, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/5bc342f48de8264779952fac378f96dc-Abstract-Conference.html.

Zihao Wang, Bin Cui, and Shaoduo Gan. SqueezeAttention: 2d management of KV-cache in LLM inference via layer-wise optimal budget. In *International Conference on Learning Representations*, volume 2025, pp. 23593–23606, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/3b0a8df568ec496a717566a7f8158aaa-Abstract-Conference.html.

Dongwei Xiao, Zhibo Liu, Yuanyuan Yuan, Qi Pang, and Shuai Wang. Metamorphic testing of deep learning compilers. *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, 6(1):1–28, 2022. doi: 10.1145/3508035. URL https://doi.org/10.1145/3508035.

Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving Mamba2 with delta rule. In *International Conference on Learning Representations*, volume 2025, pp. 29687–29707, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/4904fad153f6434a7bcf04465d4be2cc-Abstract-Conference.html.

Lu Ye, Ze Tao, Yong Huang, and Yang Li. ChunkAttention: Efficient self-attention with prefix-aware KV cache and two-phase partition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 11608–11620, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.623. URL https://aclanthology.org/2024.acl-long.623/.

Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Sun, Jeff Huang, Cody Hao Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph E. Gonzalez, Clark Barrett, and Ying Sheng. SGLang: Efficient execution of structured language model programs. In *Advances in Neural Information Processing Systems*, volume 37, pp. 62557–62583. Curran Associates, Inc., 2024. doi: 10.52202/079017-2000. URL `https://proceedings.neurips.cc/paper_files/paper/2024/hash/724be4472168f31ba1c9ac630f15dec8-Abstract-Conference.html`.

A ETHICS STATEMENT

This work introduces no human-subject data collection and evaluates only systems behavior on the PG-19 training split. It does not assess model safety, bias, privacy, or downstream deployment quality. Memory efficiency can lower deployment cost but may also increase the scale of model use; those broader effects are outside this controlled systems study.

B PROTOCOL GEOMETRY

Table 4: Frozen primary-factorial execution geometry. The eight ranks provide distinct books and query banks, not repeated stochastic measurements.

Component	Frozen value
Model	Qwen3.5-35B-A3B; 10 attention + 30 GDN layers
Hardware	8 × H20-3e; one book/rank
Software	Python 3.11.13; PyTorch 2.11.0+cu129; Transformers 5.14.1; vLLM 0.26.0+cu129; Triton 3.6.0; CUDA 12.9
KV backend	vLLM fused attention; BF16/Q16; page 128
Document / query	4,095 / 32 tokens; PG-19 train only
Generation	8 output tokens; 39 state-appended input tokens (32 query + 7 generated-token feedback)
Resident counts	$1, 8, 32$; four KV-by-GDN ownership cells
Primary schedule	all requests resident; batch-one sequential round-major calls on one CUDA stream/rank
Validation	96 factor configurations; separate memory/witness cells
Fault campaigns	$N = 2$; the primary all-gates-on campaign has nine live mutants with rebuilt controls; the target-gate-suppression campaign has nine fresh clean plus nine suppressed-gate cases on 8 H20s; the designer–executor-separated campaign has five new clean/mutant pairs on 5 H20s

Table 5: Cohort-to-claim authorization. Cohorts are never pooled; each row supports only the listed inference.

Cohort	Frozen scope	Authorized claim / explicit exclusion
Primary ownership factorial	8 books; $N = 1, 8, 32$; four KV×GDN cells; 8 decode steps	ownership lifecycle, cross-cell equivalence, selected attention oracle, live-mutant sensitivity, and four-cell allocator endpoints; not transfer or scheduler evidence
Fresh-process control	one independent $N = 2$ process; one clean path and M1	input reconstruction, cleanup, and one intended-gate reproduction; not pooled with the factorial
Lifecycle transfer	8 books; aligned 4,096-token prefix; $N = 4$; two cancelled slots	exact outputs/state, zero-scrub, exact reuse, and stale-lease rejection on the same adapter; not scheduler, concurrency, or performance evidence

Continued on next page

Table 5 continued.

Cohort	Frozen scope	Authorized claim / explicit exclusion
Scheduler interleave	8 books; $N = 4$; page/prefix/tail geometries 64/3,985/17 and 128/4,033/65; cancel slot 3 after round 2	deterministic request-step interleaving, zero-scrub, epoch change, exact reuse, uninterrupted-control equality, and three frozen scheduler-fault gates; not concurrent kernels, continuous-batching performance, detection rate, or production end-to-end correctness
Out-of-process GDN observer	one H20; fixed 4,095-token document and 32-token registered transition; two fresh $N = 2$ shared-KV cells (shared-base/materialized GDN); three phases/cell; one spawn-created observer per cell	1,080 receiver-derived descriptors and 96,660 pair relations; 6/6 phase and 2/2 lifecycle verdicts pass; no live candidate rows/verdicts or phase/policy/completion labels; producer slot enumeration and PyTorch/CUDA IPC remain trusted; not OS/driver ground truth, independent model execution, KV capture, kernel attestation, transient-write exclusion, or cross-stack evidence
Bounded two-stream lifecycle	one H20; one 4,033-token document; two 16-token query requests at $N = 2$; two host threads and distinct CUDA streams; pre-cancel and post-reclaim phases; cancellation after synchronization	positive call-interval overlap plus exact four lifecycle outputs, 20 logical-KV layer pairs, two GDN slot aggregates, ownership, scrub, and reuse; not simultaneous kernels, native continuous batching, in-flight cancellation, production scheduling, or performance
GDN transition oracle	1 frozen book/window; layers 0, 10, 20, 38; 4 query tokens	clean recurrence outputs/states and four seeded wrong-transition rejections conditional on post-normalization inputs; not all GDN layers or end-to-end semantics
Expanded captured-boundary numerical sweep	2 distinct PG-19 inputs; all 10 attention layers and 12 of 30 GDN layers; 20 attention and 24 GDN clean rows plus one seeded control/row	captured-boundary numerical agreement and rejection of all 44 seeded wrong operators; not upstream-activation, independent-capture, full-model, or end-to-end validation
Target-gate suppression matrix	9 fresh clean + 9 target-suppressed cases; same fixed model/data/$N = 2$; 8 distinct H20s	per-fault comparison with redundant audit gates, exact allowlisted production assertions, and token/FP32-logit equality; M8's injected sentinel is not a detector; not a blind-fault study or rate
Designer–executor-separated faults	separate designer reads only the prior PDF; 5 new frozen $N = 2$ clean/mutant pairs; 5 H20s; all gates enabled	all 5 clean gates pass and all 5 mutants fail first at their frozen primary predicate; per-fault sensitivity only, not a detection rate, natural-defect study, or cross-stack evidence
Mac common-mode control	Apple M4 Pro; six documents/queries; five fresh MLX processes; vanilla dense plus CoMem BF16/Q8/Q4 at depth 7	separates split-family agreement from agreement with vanilla dense; motivation only and separate from the H20 case
H20 deployment context	8 LongBench validation workloads; vanilla dense, full-prefix KV reuse, and CoMem Q8/Q4/mixed variants (six configurations total) with 3 timing repeats	same-stack reusable-state, TTFT, TPOT, tokens/s, and mean F1 comparison; not pooled with the primary factorial and not scheduler/continuous-batching evidence
Matched serving controls	8 LongBench validation workloads; vLLM 0.26 and SGLang 0.5.17 cache off/on plus 24 fixed-configuration HYPIC timing/quality cells; one H20-3e per workload	within-block cache-hit, quality, and client-wall timing context; not pooled with in-process CoMem timings, not a cross-framework leaderboard, and not continuous-batching evidence
HYPIC retained-state receipts	8 Prefix Cache + 8 HYPIC cells on the same fixed TP=1 model/slice; receipt-instrumented SGLang lifecycle	median target-entry-owned physical tensor payload only; not timing, NVML/process memory, capacity, continuous batching, or ForkAudit ownership evidence

Continued on next page

Table 5 continued.

Cohort	Frozen scope	Authorized claim / explicit exclusion
Related-work operator transfer	Hydragen at $N = 8/32$ on one captured layer-3 attention family; Palu on one layer with 8 calibration and 1 held-out PG19 row	same-checkpoint numerical and logical-state trade-offs only; not an all-layer implementation, LongBench/quality result, speedup claim, or jointly optimized system

Table 6: Trace coverage and replay verdict for the seven contract targets in the frozen case. Equality denotes canonical byte-digest identity as recorded. Overall trace coverage remains partial because dispatch lacks a per-call compiled-binary and autotuning-choice binding. A passing primary target verdict is conditional on the primary capture producer and target boundary in Section 3; the separate out-of-process GDN cohort is supporting evidence and does not independently execute the model or upgrade these seven targets to comprehensive system correctness.

Target	Trace coverage	Replay verdict	Decisive witness	Boundary
Frozen identity	complete	pass	identical preflight/terminal source ledgers, terminal model rehash, and stable data/query/protocol/execution bindings	one validated formal run; not independent producer recapture
Prefix immutability	complete	pass	physical document-KV digests and GDN base guards across all three phases	one frozen configuration/geometry
Private ownership	complete	pass	all-pairs request–base and request–peer range replay at the primary registered 32-token transition	pointer-free conservative intervals
Tail-safe append	complete	pass	one-time 127-token private-tail copy before append	one partial-tail geometry
Dispatch provenance	partial	pass at Python scope	Python callable, query/position/mask/append/scale receipts; four compiled cache artifacts bound	no per-call compiled-binary or explicit autotuning-choice binding
Cross-arm equivalence	complete	pass	tokens, shape/dtype-bound CPU-FP32 logit digests, logical KV, and GDN	one primary sequential schedule
Cross-N consistency	complete	pass	288 all-arm adjacent-fan-out comparisons over 72 rank–query identities	relational, not an independent oracle

Selected attention and GDN numerical oracles, the nine targeted live faults, the lifecycle/scheduler extensions, and the out-of-process GDN and bounded two-stream cohorts, plus the five PDF-only designer–executor-separated faults, are supporting checks rather than additional contract targets. Their captured-input, same-adapter, process-separated-but-shared-IPC, and schedule boundaries are reported separately.

C Pointer-Free Storage Witness Schema

Target-to-record dependencies. The following matrix is normative. I, L, E, and F denote the Identity, Lifecycle, Execution, and Target/fault classes in Table 16; parenthesized entries name mandatory fields. Missing any marked field makes that target open. A preregistered geometry may make tail safety not applicable; the absence of a compiled-binary fingerprint leaves dispatch partial rather than silently full. A missing F field invalidates the corresponding fault row, but does not change the separately evaluated seven-target witness.

Target	I	L	E	F	Blind predicate
Frozen identity	code, model, data, protocol; role map	—	—	N/A	exact bytes and complete roles
Prefix immutability	document/base roles; start digests	setup and final guards	canonical content digests	N/A	start = final
Private ownership	request/base/private roles	setup, registered-transition, final bindings	normalized storage ranges	N/A	non-vacuous all-pairs disjointness
Tail-safe append	page geometry; document/private roles	first append event	ordered append and copied-tail receipt	N/A	one private copy before write
Dispatch provenance	callable and protocol identity	—	ordered call, query, mask, position, append, scale	N/A	schedule and callable equality; no fallback
Cross-arm equivalence	factor, query, and schedule identities	final bindings	token, logit, logical-KV, GDN records	N/A	canonical semantic equality
Cross-N consistency	query and fan-out identities	—	the same canonical semantic records	N/A	smallest-N equals each larger N
Fault sensitivity (supporting)	frozen target/gate map	matched-control rebuild and cleanup	injected outcome and failed predicates	required	target, injection, expected gate, outcome

Each tensor-view row has the exact fields `storage_id`, `shape`, `stride`, `storage_offset`, `dtype`, `device`, `storage_nbytes`, `tensor_nbytes`, and `content_sha256`; absolute addresses are forbidden. Within each phase, the producer assigns `storage-0000`, `storage-0001`, and so on in first-observation order to backing storages. Reusing an ID with a different device or storage size is invalid. Phase capture IDs are unique, whereas request and persistent lifecycle-guard IDs must remain stable across setup, the registered transition, and final generation.

Let element size be e, storage offset o, shape s_i, and stride t_i. The independent replay initializes $m = M = o$ and, for every dimension, computes

$$d_i = (s_i - 1)t_i, \qquad m \mathrel{+}= \min(d_i, 0), \qquad M \mathrel{+}= \max(d_i, 0). \tag{3}$$

The conservative half-open byte interval is

$$I = [em,\, e(M + 1)), \tag{4}$$

which must be non-negative and contained in `storage_nbytes`; also `tensor_nbytes`$= e\prod_i s_i$. Two rows overlap iff their normalized storage IDs match and $I_a^{\text{start}} < I_b^{\text{end}}$ and $I_b^{\text{start}} < I_a^{\text{end}}$. Exact aliasing additionally requires equality of interval, geometry, dtype/device, storage and tensor sizes, and content digest. This bounding interval can conservatively flag an exotic interleaved view, but it cannot miss a shared byte.

At setup, every recurrent-state coordinate must either exactly alias the persistent state (borrowed-base arm) or be disjoint from it (materialized arm). At the primary registered 32-token transition boundary, a completed request's 60 mutable rows must be disjoint from the base and every peer; incomplete borrowed requests must remain exact aliases. At final, all request-local GDN states must be disjoint from the persistent GDN base and every peer; document KV remains intentionally shared. Independent binding-token receipts additionally require unchanged tokens for incomplete requests and changed binding/storage tokens for completed requests. The package applies this schema to all 96 timelines and 288 phase artifacts; nine adversarial tests cover exact aliasing, partial overlap, adjacent intervals, negative strides, conflicting ID reuse, pointer injection, capture/guard reuse, and cross-coordinate request–base and request–peer aliasing.

D Executed Fault Rows and Invariant Map

The row-level campaign is kept out of the main paper because it supports diagnosis and audit coverage rather than a separate headline result. Separately, the CPU governance suite passes 17 tests and rejects 28 explicit serialized-artifact tamper cases plus its registered helper/launcher faults. That result concerns schema and orchestration fail-closure only; it is not GPU semantic or performance evidence.

Table 7: Separate primary all-gates-on reference. Every separately rebuilt matched-clean cell passed; every mutant reached its target gate fixed before execution and was restored before disposal.

ID	Injected live fault	Expected/observed gate	Primary outcome
M1	reservation alias	`KV_RESERVATION_DISJOINT`	target gate reached
M2	sequence swap	`KV_SEQUENCE_ID`	target gate reached
M3	omit tail COW	`KV_TAIL_COW`	target gate reached
M4	GDN–base alias	`gdn_completed_vs_base_disjoint`	target gate reached
M5	GDN peer alias	`gdn_completed_vs_peers_disjoint`	target gate reached
M6	position off-by-one	`POSITION_CANONICAL_VALUES`	target gate reached
M7	materialized mask	`MASK_CONTRACT`	target gate reached
M8	wrong callable	`KERNEL_CALLABLE_ID`	target gate reached
M9	dense KV view	`KV_PAGED_VIEW`	target gate reached

Table 8: Representative failures each target is designed to expose. These are the nine executed primary-factorial live mutants; the checks are not complete oracles.

Target	Intended sensitivity and observed boundary
Frozen identity	wrong model/data/query bank, changed protocol, stale shard (schema adversaries)
Prefix immutability	M3 omits partial-tail COW; physical document digest and ownership gates fail
Private ownership	M1 reservation alias, M4 request–base GDN alias, M5 request–peer GDN alias
Sequence/view binding	M2 sequence swap and M9 dense-KV replacement
Dispatch provenance	M6 position offset, M7 materialized mask, M8 wrong callable
Factorial exactness	behavior changes when either KV or GDN setup ownership changes
Bounded numerical oracle	numerical error in a selected fused full-attention result
Raw replay	internally inconsistent trajectory, storage, allocator, or aggregate fields

The primary all-gates-on execution is a first-gate localization study. Every registered gate fires before the injected path produces a post-injection token, logit, logical-KV digest, or GDN digest; its output cells are therefore N/O rather than misses. Lifecycle/ownership/storage receipts supply the first gate for M1 and M3–M5; request-binding/call receipts supply it for M2 and M6–M9. The full audit rejects all nine at their predeclared gates under passing matched controls. The separate target-gate-suppression matrix then suppresses one named gate per mutant while leaving the remaining observers active. It provides the per-fault comparison in Table 1, but does not estimate a population rate, held-out-fault coverage, or false-positive rate.

E Relation to Prior Work by Audit Obligation

Table 9: Primary-source positioning. "Evidence focus" describes what the cited work evaluates; it is not a claim that the work lacks all other checks.

Work	Primary mechanism	Evidence focus / relation to ForkAudit
PagedAttention	Paged KV allocation and sharing (Kwon et al., 2023)	Serving throughput and memory utilization; supplies the block-sharing substrate.
SGLang / Prompt Cache	Prefix reuse in structured or modular prompts (Zheng et al., 2024; Gim et al., 2024)	Runtime performance and output behavior; establish reuse and positional handling as prior art.
CoMem	One persistent residual per token at split j; bounded selection; suffix replay (Liu et al., 2026a;b)	Motivates the reusable document object; the bounded deployment table supplies same-stack context, while the contribution here remains the hybrid-state ownership audit.
Marconi	Admission and eviction for hybrid-model prefixes (Pan et al., 2025)	Hybrid cache hit rate and time to first token; closest state-heterogeneous caching system, but a different policy-level contribution.
SCBench	Lifecycle-oriented KV-cache benchmark (Li et al., 2025)	Breadth over tasks and cache phases; exposes breadth absent from our single reference case.
MT-DLComp / CRADLE	Metamorphic compiler testing and cross-backend differential checking (Xiao et al., 2022; Pham et al., 2019)	Direct testing precedents; our audit adds ownership witnesses and a layerwise dense oracle, but not an independent end-to-end backend.
DNNMem	Component-wise GPU-memory accounting (Gao et al., 2020)	Supports denominator separation; it is an estimator, not a request-ownership audit.
ForkAudit	Typed, phase-indexed ownership/call/accounting trace with explicit coverage semantics	Integrates a mutation-tested KV-by-GDN ownership factorial, captured-boundary attention/GDN numerical sweeps, same-adapter lifecycle/scheduler-contract checks, and bounded out-of-process GDN storage observation; no new cache policy, production scheduler, OS/driver allocation monitor, independent model execution, or end-to-end oracle.

Table 10: Closest established precedent for each audit obligation. The final column states the narrower integration contributed by ForkAudit; it does not claim that any individual mechanism is new.

Audit obligation	Closest established precedent	Integrated obligation in this work
Frozen identity	Controlled variants in metamorphic and differential testing (Chen et al., 2018; Xiao et al., 2022; Pham et al., 2019)	Bind model, data, query bank, callable contract, and protocol bytes to every ownership receipt.
Document immutability and tail COW	PagedAttention block sharing and copy-on-write for shared prefixes (Kwon et al., 2023)	Replay physical document pages, padding, active tables, and the partial-tail detach event.
Recurrent-state ownership	Hybrid recurrent states and their in-place update constraint (Yang et al., 2025; Pan et al., 2025)	Witness request–base and request–peer ranges before and after the frozen schedule's registered transition boundary.
Dispatch provenance	Cross-backend differential checking exposes implementation-dependent behavior (Pham et al., 2019)	Hold the backend fixed and record callable, position values, mask, scale, and append history per call.
Cross-arm and cross-N relations	Metamorphic relations over semantics-preserving transformations (Chen et al., 2018; Xiao et al., 2022)	Treat ownership policy and resident fan-out as transformations that must preserve registered request observables.
Bounded numerical oracle	Independent backend comparison (Pham et al., 2019)	Reconstruct selected attention inputs for dense IEEE-FP32 math and selected post-normalization GDN inputs for a candidate-import-free NumPy recurrence.
Memory denominators	Component-wise GPU-memory accounting (Gao et al., 2020)	Keep page geometry, allocator current/peak, and unmeasured process/service capacity as separate claim domains.

Table 11: Unpooled Mac motivation context (Apple M4 Pro, five fresh MLX processes; median over 30 query executions at split depth 7). Vanilla dense is the external reference. CoMem BF16/Q8/Q4 variants use the same selected documents. Store size is the six-document residual corpus; online time includes query Write, dequantization, and suffix Read. This small teacher-forced demonstration set is not primary ForkAudit evidence and supports no cross-hardware or production-speed claim.

Method	Store (MiB)	Online (ms)	Speedup	Top-1 vs. dense
Vanilla dense prompt	–	194.70	1.00×	100.00%
CoMem BF16 split	1.998	168.88	1.15×	71.89%
CoMem Q8 split	1.061	169.17	1.15×	71.89%
CoMem Q4 split	0.562	169.25	1.15×	71.89%

Q8/Q4 retain 100% top-1 agreement with the BF16 split path, yet all split rows inherit only 71.89% agreement with vanilla dense. The speed column is a small teacher-forced mechanism benchmark, not production RAG throughput.

F Supporting Control Experiments

Mac common-mode motivation. The archival Apple-M4-Pro control is kept unpooled from the H20 experiments and serves only to show why split-family agreement does not replace a dense or independently implemented oracle.

Published systems context. Table 12 reports each cited system only in its native protocol and keeps all non-commensurate metrics outside our matched panels. This unpooled table is literature context, not ForkAudit evidence.

Table 12: Unpooled published-systems context in each paper's native protocol. These values show the scale and evaluation breadth of related systems, *not* a cross-paper leaderboard or ForkAudit evidence: models, hardware, batching, baselines, and metrics differ. Consequently, no row is merged with our same-stack H20 Table 3, and the absence of a common score must not be read as a negative result.

System	Native method and study setting	Reported efficiency in its native protocol	Quality and comparability boundary
PagedAttention / vLLM (Kwon et al., 2023)	Paged KV allocation plus serving engine *Setting:* LLaMA-7B on A10G and LLaMA-13B on A100-40GB; vLLM serving engine	2–4× serving throughput at similar latency vs. FasterTransformer/Orca	No common LongBench F1; exact cache-management method
Prompt Cache (Gim et al., 2024)	Modular attention-state reuse *Setting:* Llama2/MPT/Falcon; RTX 4090, A40, and A100; HuggingFace-style prototype	1.5–3.1× TTFT (CPU-resident modules); 3.7–11.7× (GPU-resident) on its 12-dataset LongBench suite	LongBench accuracy evaluated, but with different models and hardware
SGLang / RadixAttention (Zheng et al., 2024)	Prefix reuse plus structured-program runtime *Setting:* Llama-7B/Mixtral-8x7B and structured programs; A10G/A100; SGLang	Up to 6.4× throughput vs. prior inference systems	Application-suite result; no common LongBench F1
ChunkAttention (Ye et al., 2024)	Prefix-aware KV chunks and attention kernel *Setting:* attention microbenchmarks; mainly A100-80GB, plus A100-40GB/RTX 4090; CUDA 11.8	3.2–4.8× attention-kernel speedup for 1K–4K shared prompts	Kernel result; no end-to-end common task score
Hydragen (Juravsky et al., 2024)	Exact batched shared-prefix attention *Setting:* CodeLlama-13B/34B shared-prefix batches; A100 (with H100/L40S microbenchmarks); custom HuggingFace runtime	Up to 32× CodeLlama-13B end-to-end throughput in large shared-prefix batches	Exact attention construction; different model and batching regime
Palu (Chang et al., 2025)	Low-rank latent KV with reconstruction/fusion *Setting:* Llama/Mistral/LongChat quality studies; Llama-2-7B latency on one RTX 4090; HuggingFace plus custom kernels	50% KV compression and up to 1.89× RoPE-attention speedup; up to 2.91× with quantization	Llama/Mistral/LongChat native study; our bounded Qwen3.5 operator transfer is reported separately
Preble (Srivatsa et al., 2025)	Distributed prompt-sharing scheduling *Setting:* Mistral-7B/Llama-3-70B request traces; two–eight GPUs across A6000/H100 clusters; distributed serving	1.5–14.5× average-latency and 2–10× p99-latency improvement	Distributed serving traces; no common task score
Marconi (Pan et al., 2025)	Admission/eviction for hybrid-model prefixes *Setting:* Attention–Mamba2 7B geometry; LMSys/ShareGPT/SWE-bench traces; CloudLab policy simulation	Up to 34.4× token-hit rate and up to 71.1% or 617 ms lower TTFT	Hybrid models and trace-driven policy study; no common LongBench F1
HYPIC (Liu et al., 2026c)	Position-independent caching for hybrid full/linear-attention models *Setting:* Qwen3.5-35B-A3B main panels at TP=2 on 8 H20-3e; SGLang 0.5.14	3.25× lower TTFT on average and 1.66× higher QPS over Prefix Cache	Four models/five workloads; our same-slice TP=1 timing and retained-state controls are reported separately

Same-checkpoint transfer detail. Table 13 is the matched complement to the published context table: it uses our frozen Qwen3.5 checkpoint and H20-3e, binds the official Hydragen and Palu source revisions, and independently replays all raw operator sidecars. Because it does not execute all layers or LongBench, it is kept separate from Table 3.

Table 13: Bounded same-checkpoint H20 transfers. Hydragen uses frozen Qwen3.5 layer-3 post-RoPE attention; Palu uses eight 256-token PG19 calibration rows and one disjoint held-out row. Errors are relative ℓ_2 against the matched uncompressed operator; state ratios are logical selected-operator ratios, not process memory or end-to-end results.

Port	Setting	Matched numerical result	Logical state result	Timing / boundary
Hydragen port	$N = 8$	Hydragen–dense rel. ℓ_2 = 0.00255	8.50 vs. 64.48 MiB (7.59× smaller)	0.287 vs. 0.0965 ms; no speedup
Hydragen port	$N = 32$	Hydragen–dense rel. ℓ_2 = 0.00262	10.00 vs. 257.94 MiB (25.80× smaller)	0.290 vs. 0.259 ms; no speedup
Palu port	rank 64/head	held-out K/V 0.487/0.549 (plain 0.711/0.839)	512 vs. 2,048 B/token (4.00× smaller)	projection only; no fused kernel
Palu port	rank 128/head	held-out K/V 0.317/0.364 (plain 0.543/0.668)	1,024 vs. 2,048 B/token (2.00× smaller)	projection only; no fused kernel
Palu port	rank 192/head	held-out K/V 0.149/0.194 (plain 0.341/0.459)	1,536 vs. 2,048 B/token (1.33× smaller)	projection only; no fused kernel

Hydragen avoids prefix replication but has no speedup here. Palu improves over plain SVD, but residual error forbids an end-to-end quality claim. Neither is a jointly optimized combined system.

Matched serving-framework and HYPIC context. Table 14 gives separately timed vLLM and SGLang cache-off/on controls plus the official-code HYPIC Full Recompute, Prefix Cache, and full HYPIC arms under the shared Qwen3.5 streaming boundary. It remains a timing/quality-only panel and authorizes only within-framework interpretation; the separate 16-cell retained-state receipt cohort appears in Table 3. vLLM and SGLang each record 8/8 prefix hits and preserve 8/8 cache-off predictions. Within its TP=1 block, HYPIC gives 39.63, 0.101 s, and 17.67 tokens/s; published Qwen3.5 panels use TP=2, so this is an adaptation. Its 24 timing cells bind a 4,404-entry terminal artifact ledger. Across the 16 retained-state cells, Prefix Cache retains a median 139.53 MiB and HYPIC retains 324.09 MiB; CoMem Q8 is 8.78× and 20.39× smaller, respectively, under the separate Store estimand. This remains single-stream; it does not measure continuous batching.

Table 14: Unpooled timing/quality-only serving context on Qwen3.5 and the same eight-item Qasper/2WikiMQA slice. Each row uses one H20-3e per item, a 4,096-token cap, greedy decoding, and at most 32 generated tokens. Times are OpenAI-compatible streaming client wall-clock; F1 is the eight-item mean. Hit is interpreted only against the disabled/full-recompute reference inside each framework block. No cross-framework speedup, retained-state, scheduler-throughput, capacity, or broad-quality claim is made.

Framework	Prefix reuse	F1	TTFT (s)	TPOT (ms)	tok/s	Hit
vLLM 0.26	off	39.63	1.445	56.31	4.70	–
vLLM 0.26	align on	39.63	0.179	53.99	14.00	8/8
SGLang 0.5.17	off	39.14	0.283	53.38	12.73	–
SGLang 0.5.17	Radix on	39.14	0.146	53.15	15.83	8/8
HYPIC code	full recompute	39.14	0.217	49.21	14.77	–
HYPIC code	prefix cache	39.14	0.072	50.15	19.07	8/8
HYPIC	transition+seam 8	39.63	0.101	52.13	17.67	8/8

vLLM uses experimental Mamba `align`; SGLang 0.5.17 uses `extra_buffer` Radix caching. The HYPIC rows use the authors' commit 98147c0 on SGLang 0.5.14, TP=1, full `transition_rope_recompute`, and an eight-token seam. Its full and prefix rows are controls from that same codebase; published TP=2 results and the separate retained-state receipt cohort are not inserted here.

Hybrid-prefix policy context. The separate preregistered Marconi trace extends the eight frozen workloads with a fixed synthetic follow-up turn and evaluates only prefix-policy token reuse under the artifact's native Attention–Mamba2 geometry. It neither reruns LongBench quality nor supplies Qwen3.5 serving speed, so it is not pooled with the preceding panel or the primary audit.

Table 15: Unpooled preregistered policy-simulator context on a 128-event synthetic multi-turn extension of the eight frozen workloads. Values are token-hit percentages under the official Marconi artifact's native Attention–Mamba2 geometry. The wrapper calls each policy's own eviction routine after insertion to close actual tree size to the declared budget. These are not Qwen3.5 quality, latency, or throughput measurements and do not validate the primary ownership audit.

Policy	5 GB	10 GB	20 GB
vLLM+	2.07	6.18	46.83
SGLang+	90.78	93.86	93.86
Marconi	**93.86**	93.86	93.86

Table 16: Normative record classes. "Mandatory" means that absence makes the corresponding target open rather than silently not applicable.

Record class	Mandatory payload	Independent replay
Identity	frozen code, model, data, protocol, and object-to-role map	exact bytes and role coverage
Lifecycle	setup, registered-transition, and final inventories and bindings	normalized ranges, aliases, and disjointness
Execution	ordered call/append receipts, semantic records, named allocator snapshots	frozen schedule, digests, and phase equations
Target/fault	predeclared target, gate, matched control, injection, and outcome	failed-predicate set and classified outcome

Primary allocator endpoints. Table 17 keeps the four ownership cells and their allocator denominators together under the primary 8-rank cohort.

Table 17: Primary 8×H20-3e allocator results at $N = 32$ (median across ranks; rank values coincide). Vanilla full-copy is the external control; the paged-prefix row is the closest same-stack prefix-sharing baseline. The remaining rows isolate recurrent-base ownership. All cells retain the persistent document source. Values are GiB above the frozen post-priming baseline; generation increment is the additional peak above allocation present at generation start.

Empirical arm	KV / GDN setup	Final	Peak	Gen. incr.
Vanilla full-copy	Full-copy KV / Materialized GDN base	4.901	4.920	0.019
GDN ownership ablation	Full-copy KV / Borrowed GDN base	4.890	4.907	1.951
Paged-prefix baseline	Shared-doc KV / Materialized GDN base	2.229	2.843	0.019
Audited hybrid fork	Shared-doc KV / Borrowed GDN base	2.229	2.843	1.950

Aligned cancellation and reclamation lifecycle. The separate lifecycle-transfer cohort uses eight ranks, eight distinct PG-19 training books, an aligned 4,096-token prefix, and $N = 4$. It is not pooled with the primary factorial.

Table 18: Lifecycle-transfer obligations on the existing Qwen3.5/vLLM Q16 adapter. Counts are deterministic rank outcomes, not stochastic replicates.

Obligation	Result
Frozen static/code/model/weight/raw bindings	pass; 14/14 weights, 8/8 shards
Full-vocabulary logits and generated tokens vs. uninterrupted control	exact, 8/8
Final logical KV and final GDN state	exact, 8/8
Aligned prefix, immutable document, and zero partial-tail copy	pass, 8/8
Cancelled-slot zero-scrub and exact reservation reassignment	pass, 8/8
Stale epoch-0 rejection and matched epoch-1 acceptance	8/8; 0 escapes, 0 wrong gates

Scheduler-managed request-step interleaving. The separate formal extension keeps four requests resident, cancels slot 3 after its second round, zero-scrubs and increments its lease epoch, reassigns the exact reservation, and completes the replacement interleaved with the three survivors. It is not pooled with the primary factorial or the aligned lifecycle-transfer cohort.

Table 19: Formal scheduler-managed request-step interleaving on the frozen Qwen3.5/ForkAudit vLLM-Q16 stack. Clean counts are deterministic rank outcomes; fault counts are preregistered expected-gate outcomes, not a detection rate.

Case	Frozen geometry or intervention	Outcome
Clean geometry 1	page 64; 3,985-token prefix; 17-token document tail; 18-event cancel/replacement lifecycle	all semantic/storage gates and uninterrupted-control comparisons pass, 8/8
Clean geometry 2	page 128; 4,033-token prefix; 65-token document tail; same lifecycle	all semantic/storage gates and uninterrupted-control comparisons pass, 8/8
FH1	dispatch cancelled epoch-0 lease	`STALE_SLOT_LEASE`, 16/16
FH2	dispatch lease under another request	`LEASE_REQUEST_MISMATCH`, 16/16
FH3	reclaim before zero scrub	`RECLAIM_NOT_ZERO`, 16/16

Candidate-import-free GDN transition oracle. The oracle uses one frozen PG-19 window and four query-time layer transitions at layers 0, 10, 20, and 38. A single preregistration frozen before the successful run binds the post-native-q/k-normalization boundary, scale $1/\sqrt{128}$, tolerances, and seeded faults. Its NumPy reference starts at that captured boundary; no candidate module is imported by the reference.

Table 20: PDF-visible oracle selection and replay boundary. Every listed row was locked before its successful candidate execution; neither oracle validates upstream activation construction or end-to-end model behavior.

Oracle	Frozen rows / selection rule	Captured boundary	Decision rule	Replay availability
Attention	rank r uses layer $4r + 3$ and round r, for $r \in \{0, \ldots, 7\}$; request 0, $N = 1$, shared-document KV + borrowed GDN; frozen rank/layer/round map	post-RoPE Q/K/V, positions, mask, scale, and ordered append shadows	dense CPU-FP32 rel. $L_2 \leq 0.005$	8 raw rows and numerical sidecars; primary-package one-command replay
GDN	layers $0, 10, 20, 38$; four query tokens from one frozen PG-19 window	post-native-q/k-normalization recurrence; scale $1/\sqrt{128}$	output rel. $L_2 \leq 0.005$; state rel. $L_2 \leq 10^{-4}$; seeded fault must reject	4 clean + 4 fault decisions from 40 sidecars; separate one-command replay

Table 21: Bounded GDN recurrence oracle. Relative errors compare native outputs and final states with the candidate-import-free NumPy FP32 recurrence.

Layer	Output rel. L_2	State rel. L_2	Clean	Seeded fault	Rejected
0	0.001584	1.23×10^{-7}	yes	omit decay	yes
10	0.001652	1.36×10^{-7}	yes	complement β	yes
20	0.001572	1.39×10^{-7}	yes	pre-decay memory	yes
38	0.001638	1.30×10^{-7}	yes	roll value heads	yes

Table 22: Preregistered expanded captured-boundary numerical sweep. Clean rows use candidate-import-free CPU/NumPy FP32 replay; every clean row has one seeded wrong-operator control. Attention begins after RoPE and GDN begins after native q/k normalization, so the table does not validate upstream activation construction, independent capture, or end-to-end model behavior.

Operator	Inputs	Layers	Clean rows	Positions / transitions	Max output rel. L_2	Controls rejected
Attention	2	10/10	20/20	160 positions	0.001897	20/20
GDN	2	12/30	24/24	192 transitions	0.002073 (2.08×10^{-7} state)	24/24

Full target-gate-suppression matrix. The preregistered target-gate-suppression execution contains all M1–M9 faults and a fresh matched clean path for each. Across eight distinct H20s, all 18 cases complete their declared horizon or an allowed classified stop, all 24 FP32 sidecars verify, and no case is operationally invalid. Five mutant paths complete semantics: token equality catches 0/5, exact logits catch M5 only ($L_\infty = 0.75$, relative $L_2 = 0.0314$), and M1/M2/M6/M7 leave both exact. M3/M4 reach other audit gates; M9 reaches the paired-view production assertion; M8 reaches the exact injected payload sentinel, which is not a production detector. The frozen aggregator initially read the inherited execution identifier from a detached-manifest field that the manifest schema omits. A disclosed postexecution wrapper derives that identifier from the canonical receipt embedded in all eight manifest-bound primary-package shards, changes only the generated comparison-row `run_id` from null to the preregistered value, and reproduces the summary byte-for-byte without modifying candidate outputs or frozen sources.

G Expanded Limitations and Claim Boundary

The primary factorial covers one model, one H20 hardware family, one page size, 16-bit KV, and $N \leq 32$. Its ownership conclusion is tied to one 4,095-token partial-tail geometry, a registered 32-token continuation block, and the subsequent frozen sequential schedule; it is not a claim about an arbitrary first token or transition length. The lifecycle cohort adds one aligned 4,096-token geometry and a fixed cancel/scrub/reclaim sequence. The scheduler-interleave extension adds two geometries and deterministic request-step interleaving, but those requests execute sequentially on one CUDA stream.

The separate two-stream cohort has two host workers and distinct CUDA streams with positive full-model-call interval overlap. Its intervals include queuing, resource waits, and host-enqueue gaps; there is no per-kernel overlap trace. Cancellation occurs only after synchronization, so the cohort does not test arbitrary or in-flight reclamation. The out-of-process GDN observer removes same-process reconstruction and live judgment labels, but the producer still enumerates and semantically binds the frozen slots; both sides trust PyTorch tensor/storage and CUDA-IPC semantics, and mutation is paused for each snapshot. It does not supply OS/driver allocation ground truth, independent model execution, KV/kernel attestation, or protection against a transient write followed by restoration.

No cohort evaluates native continuous/ragged batching, eviction, multi-document serving, general scheduler cancellation, process-level NVML memory, OOM capacity, or a broad downstream benchmark. Full-copy KV and materialized GDN bases are controlled ownership factors rather than optimized production baselines. The GDN path is the Transformers Torch implementation, not an optimized recurrent kernel. The compiled-cache ledger does not bind each call to a selected binary or expose an explicit autotuning artifact.

The auxiliary serving panel uses one request stream per engine and compares each cache-enabled framework only with its own disabled reference. It does not measure continuous batching, eviction, admission control, maximum concurrency, or best-tuned performance. CUDA graphs were disabled in both pinned runs; the reported values describe that fixed configuration only.

The HYPIC timing/quality comparison uses the authors' exact tracked commit on SGLang 0.5.14 in one fixed CUDA 12.9 / PyTorch 2.11 configuration, adapting Qwen3.5 execution from the paper's TP=2 to TP=1 so every arm uses one H20-3e per workload. Cache-hit authority comes from OpenAI response usage. Each of its 24 timing/quality cells has one formal measurement after a discarded, prefix-disjoint warmup. A separate receipt-instrumented run uses the same model/slice and fixed TP=1 serving configuration to produce 8 Prefix Cache and 8 HYPIC retained-state cells; each completed cell binds raw output plus pre-measurement and terminal ownership receipts, and all 16 pass an external hash-bound replay. The instrumentation adds target-entry byte-range receipts and a lifecycle-idempotency overlay; Store therefore does not come from the unmodified timing run. Neither cohort establishes scheduler throughput, QPS, capacity, the paper's TP=2 latency, or accuracy robustness beyond this eight-item slice.

The same-checkpoint related-work transfer is likewise operator-bounded: Hydragen uses one captured layer-3 shared-prefix attention family, and Palu uses one layer with eight calibration rows and one held-out row. Neither executes all model layers, a fused Palu kernel, LongBench, or a jointly optimized Hydragen/Palu-plus-CoMem serving path.

The page equations are deterministic identities, not learned laws; there is one observation per rank–configuration cell and no stochastic replication. Most runtime tensors are represented by hashes and equality receipts rather than archived in full. The attention and GDN oracle rows are the exception and ship numerical sidecars, but cover only selected captured-input operator boundaries rather than an end-to-end model. The expanded numerical sweep covers two short inputs, all ten attention layers, and 12 of 30 GDN layers; it does not establish length/input robustness or full recurrent-layer coverage. Both the original and expanded GDN runs start after native q/k normalization and therefore do not validate that upstream boundary. Pointer-free normalized intervals plus raw-byte hashes make the primary captured storage evidence replayable. The out-of-process observer reconstructs receiver-visible GDN descriptors and relations in another process, but still trusts producer-side slot selection and the IPC substrate; neither constitutes protection against a malicious producer before capture. The nine mutants are targeted positive controls; the target-suppression matrix adds a same-system comparison for those same designed faults, not blind or naturally occurring defects. Four complete with unchanged measured tokens and exact logits, one changes only full logits, and four stop earlier; N/O cells are not misses, while the injected M8 sentinel is not a production detector. This is not a general detection rate, a blind-fault result, or superiority over every conventional test suite. The scheduler extension's three frozen faults are expected-gate outcomes, not a detection rate or evidence of fault-set completeness. Broader systems claims would additionally require a native production serving scheduler, per-kernel overlap evidence, in-flight cancellation tests, optimized controls, alignment/length/input sweeps, and process-level capacity accounting.

The five additional designer–executor-separated faults are genuinely fresh relative to the disclosed PDF fault list and were fixed before executor preparation, but they remain deliberately constructed, fixed-stack mutations. Passing all five does not estimate unseen-fault recall, false-positive rate, or robustness to arbitrary runtime, model, schedule, or adversarial capture defects.

H Artifact and Claim Map

Artifact footprint. The source-complete primary package, derived summaries, raw ledgers, and expanded numerical sidecars are content-bound and locally replayable. The table below maps each manuscript claim to its machine-readable source and validation boundary; supporting measurements not used by a manuscript claim remain in the internal experiment registry.

Table 23: Artifact and claim map. Relative paths are rooted at the accompanying research package. Completed rows authorize only their stated bounded claims. Metadata-redacted derivatives preserve numerical fields and state digests, while most full tensors still require the pinned environment for independent numerical recomputation.

Claim/evidence	Machine-readable source
Primary factorial replay package	`evidence/round_04_rr2_package/`
Replay and package manifest	`evidence/round_04_rr2_package/replay/run_replay.sh` and `evidence/round_04_rr2_package/MANIFEST.json`
Registered, derived, and raw evidence	`upstream/forkaudit-summary.json`, `derived/derived_summary_v2.json`, and `upstream/raw/` under the primary replay package
Executed source and storage tests	`executed_source/gpu/` and `replay/test_storage_witness.py` under the primary replay package
Identity, lifecycle, scheduler, and GDN extensions	`evidence/rr2_terminal_closure_a1_20260821/`, `evidence/lifecycle_transfer_reviewer_package/`, `evidence/round23_a2_scheduler_interleave/`, and `evidence/gdn_transition_oracle_preregistered_20260819d/`
Mac and H20 deployment controls	`evidence/mac_m4_motivation/`, `evidence/h20_deployment_benchmark/`, and their generators under `scripts/`
HYPIC retained-state receipts	`evidence/related_work_same_protocol/hypic-retained-state-r34-trial1892234/`
Target-gate-suppression preregistration and raw ledger	`evidence/r28_full_detector_matrix/formal_run_20260824a/preregistration/detector-matrix-v2-plan.json` `evidence/r28_full_detector_matrix/formal_run_20260824a/receipts/raw-artifacts.sha256`
Target-gate-suppression summary and replay receipt	`evidence/r28_full_detector_matrix/formal_run_20260824a/detector-matrix-v2-summary.postexec-corrected.json` `evidence/r28_full_detector_matrix/formal_run_20260824a/postexecution-correction-replay-receipt.json`; disclosed field-source amendment: `evidence/r28_full_detector_matrix/postexecution-rr2-run-binding-correction.json`
vLLM, SGLang, HYPIC timing/quality, and Marconi controls	`evidence/related_work_same_protocol/`; exact package members and one-command replays reside under this directory
Hydragen/Palu same-checkpoint transfers	`evidence/round24_related_work_transfer/` and its `executed_source/` directory
Published context and table generator	`literature/reported_system_context.json` and `scripts/generate_related_work_context_table.py`
Out-of-process GDN observer	`evidence/r33_independent_capture/formal_h20/result/preregistration/preregistration.json` `evidence/r33_independent_capture/formal_h20/result/raw/out-of-process-gdn-capture.json` `evidence/r33_independent_capture/formal_h20/result/replay/out-of-process-gdn-replay.json` `evidence/r33_independent_capture/formal_h20/result/receipts/terminal-files.sha256`

Continued on next page

Table 23 continued.

Claim/evidence	Machine-readable source
Designer–executor-separated faults	`evidence/r33_fresh_faults/author_freeze/designer_input/round32_input.pdf` `evidence/r33_fresh_faults/author_freeze/PROTOCOL.md` `evidence/r33_fresh_faults/executor_attempt_b/AMENDMENT.md` `evidence/r33_fresh_faults/formal_h20/RESULT_VERIFICATION.json` `evidence/r33_fresh_faults/formal_h20/r33-fresh-faults-20260825b/rank-run-*/rank-*/mutant-case.json` `evidence/r33_fresh_faults/formal_h20/r33-fresh-faults-20260825b/summary.json` `evidence/r33_fresh_faults/formal_h20/r33-fresh-faults-20260825b/terminal-files.sha256`
Bounded two-stream lifecycle	`evidence/r29_true_concurrency/formal_run_20260825b/raw/formal-result.json` `evidence/r29_true_concurrency/formal_run_20260825b/replay/independent-replay.json` `evidence/r29_true_concurrency/formal_run_20260825b/receipts/raw-and-replay.sha256`
Expanded captured-boundary numerical sweep	`evidence/r30_expanded_oracle_sweep/`; 272 local arrays under `raw/sidecars/`; canonical result `oracle-result.json`; local replay source `r30_expanded_oracle_reference.py`; validation `validation_report.json`; terminal binding `terminal-products.sha256`
Manuscript claim index	`evidence/claim_evidence_map.tsv`